Fibrewise Born–Infeld Capacity Lapse and Gravitational Time Dilation

(LC-O2 of the Lorentz Capacity Programme)

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2026

Abstract

The note [3] identifies the construction of a temporal residual map $\mathcal{P}_\tau^{\mathrm{Q5b}}$ as the remaining step needed to close the Lorentz identification from O7 observables. The paper [4] carries out this construction in the homogeneous regime and establishes the uniqueness of the resulting map. The present paper extends this construction to the non-homogeneous fibrewise regime; the homogeneous case is recovered as Corollary 4.3.

In a non-homogeneous configuration a localised spectral occupancy $\epsilon(x)$ reduces the available BI capacity budget at each point x to a local radius $R(x)^2 = 1 - \epsilon(x)$. The Weil filling law and the Lorentzian splitting are unchanged. A fibre-by-fibre uniqueness argument forces the *fibrewise* temporal residual map:

$$\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\Sigma^{\mathrm{tot}}(x,n) \;=\; R(x)\,\sqrt{f(x,n)\big(2-f(x,n)\big)},$$

where $f(x,n) = 1-\beta(x,n)$ and $\beta(x,n)$ is the local normalised spatial mobility. Gravitational time dilation is identified as the position-dependence of the temporal residual capacity:

$$\left.\frac{d\tau}{dn}\right|_x = R(x)\,\sqrt{1-\beta^2} = \frac{R(x)}{\gamma}.$$

The local capacity radius is identified metrically as $R(x)^2 = -2\,g_{\tau\tau}(x)$, where $g_{\tau\tau}(x)$ is the inverse metric temporal component and the factor of 2 comes from the flat-space reference $g^\infty_{\tau\tau} = -1/2$ (Q11 [5]). In the weak-field regime, the Newtonian profile of $g_{\tau\tau}(x)$ established in [1] gives $R(x) \approx 1 + \Phi_N(x)/c^2$, recovering gravitational time dilation as a consequence of the local BI capacity budget. The open spectral problem is to derive $R(x)$ directly from the local occupancy $\epsilon(x)$ without importing the Einstein equations (LC-O2-O1).

The purpose of this note is not to rederive the Einstein equation. The infrared Einstein response is supplied by the renormalized spectral entropy functional of [1]. The purpose is instead to identify the local kinematic meaning of gravitational time dilation in the capacity-flow language: a non-uniform projected configuration reduces the local Born–Infeld capacity radius, and the temporal residual map becomes fibrewise. In summary: special-relativistic time dilation is the homogeneous capacity trade-off between spatial mobility and temporal residual capacity; gravitational time dilation is the fibrewise version of the same trade-off, in which the available capacity radius is reduced by local spectral occupancy.

Keywords. fibrewise Born–Infeld capacity, gravitational time dilation, capacity lapse, local capacity radius, temporal residual map, Lorentz factor, non-homogeneous effective geometry, emergent gravitation, Cosmochrony.

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1 Introduction

1.1 From homogeneous to fibrewise

The note [3] establishes, conditionally on the construction of $\mathcal{P}_\tau^{\mathrm{Q5b}}$, that the core identity

$$\mathcal{P}_\tau^{\mathrm{Q5b}}\Sigma_n^{\mathrm{tot}} = \sqrt{1 - B_n^2} \tag{1}$$

recovers $\Phi(\eta_n^\tau) = 1/\gamma$. The paper [4] closes this construction unconditionally and proves uniqueness of the homogeneous temporal residual map. The present paper extends this to the fibrewise non-homogeneous regime; the homogeneous case ($R(x) \equiv 1$) is recovered as Corollary 4.3. Gravitational time dilation, in this language, asks what happens when condition (BI) is modified: when the BI capacity budget is no longer a unit sphere everywhere but a sphere of locally varying radius $R(x) \leq 1$.

This is the question addressed in the present paper. We do not ask how the local budget variation $\epsilon(x) = 1 - R(x)^2$ arises dynamically; this is the domain of the renormalized spectral entropy functional of [1], which produces the Einstein tensor as the infrared-dominant response to a metric-dependent spectral operator. We ask instead: given a position-dependent local capacity radius $R(x)$, what is the fibrewise extension of the temporal residual map, and what kinematic identity does it produce?

The answer is structurally forced by the same uniqueness argument as in the homogeneous case, applied fibrewise.

1.2 Two choices and the canonical one

When the BI budget becomes local, there are two natural ways to extend the observables.

Absolute capacity formulation. Keep $B(x, n)$ as an absolute fraction of capacity and replace the unit sphere by $\big(F^\tau(x, n)\big)^2 + B(x, n)^2 = R(x)^2$. The temporal residual is then $F^\tau(x, n) = \sqrt{R(x)^2 - (1 - f(x, n))^2}$, defined only when $1 - f(x, n) \leq R(x)$. This mixes the local budget into the definition of $B(x, n)$ and loses the clean separation between Lorentzian geometry and local scale.

Normalised capacity formulation (canonical). Write $B(x, n) = R(x)\,\beta(x, n)$ with $\beta(x, n) \in [0, 1]$ the local normalised spatial mobility, and $f(x, n) = 1-\beta(x, n)$. The BI sphere in normalised variables is $(F^\tau)^2/R(x)^2 + \beta^2 = 1$, giving

$$F^\tau(x, n) = R(x)\,\sqrt{1 - \beta(x, n)^2}. \tag{2}$$

The temporal residual map then factorises as

$$\mathcal{P}_{\tau,x}^{\mathrm{Q5b}} \quad \longleftrightarrow \quad \underbrace{R(x)}_{\text{local budget}} \quad \times \quad \underbrace{\sqrt{f(x, n)(2 - f(x, n))}}_{\text{normalised Lorentzian geometry}} \quad . \tag{3}$$

This is the canonical formulation: $R(x)$ carries all the local physics, while $\sqrt{f(2 - f)}$ is the universal normalised Lorentzian structure established in [4].

1.3 Conceptual interpretation

Capacity arbitration. The Born–Infeld budget is a local capacity arbitration principle. In the homogeneous case, spatial mobility consumes part of the unit projective capacity budget, leaving a temporal residual $1/\gamma$. In the non-homogeneous case, a local projected configuration has already depleted part of the available capacity, reducing the local radius of the BI sphere

to $R(x)$. Thus kinematic time dilation and gravitational time dilation are two independent manifestations of the same capacity trade-off:

$$\left.\frac{d\tau}{dn}\right|_x = R(x)\,\frac{1}{\gamma}. \tag{4}$$

The factor $1/\gamma$ measures the capacity diverted into spatial mobility, while $R(x)$ measures the capacity already occupied by the local spectral configuration. In this reading, curvature is the effective metric expression of local projective-capacity depletion.

Pivot statement. Special-relativistic time dilation is the homogeneous capacity trade-off between spatial mobility and temporal residual capacity. Gravitational time dilation is the fibrewise version of the same trade-off, in which the available capacity radius is reduced by local spectral occupancy.

Epistemic boundary. The present result does not derive the Einstein equations and does not yet derive $R(x)$ directly from admissibility data alone. It gives the fibrewise capacity interpretation of the metric lapse once the effective metric has been supplied by the infrared Einstein response of [1]. The remaining open problem is to prove, within the capacity-flow language itself, that

$$\epsilon(x) = 1 - R(x)^2 = 1 + 2g_{\tau\tau}(x).$$

Until this bridge is closed, the equivalence between spectral occupancy and metric lapse is an imported Einstein-regime identification, not an internal derivation.

1.4 Main results and organisation

Section 2 fixes the fibrewise extension of the capacity bundle. Section 3 defines $R(x)$ and identifies it with the metric. Section 4 states and proves the main theorem (fibrewise map) and the homogeneous corollary. Section 5 extracts the gravitational time dilation formula. Section 6 derives the weak-field limit from [1]. Section 7 gives the logical status table. The conceptual interpretation and epistemic boundary are in Section 1.3.

2 Fibrewise Capacity Bundle

2.1 Notation

Throughout, (M, g) denotes the effective Lorentzian spacetime $M = \mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ with completed co-metric $g^{\mu\nu}(x)$. In the homogeneous regime $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Q11 [5]). In the non-homogeneous regime, $g^{\mu\nu}(x)$ is the effective co-metric determined by the infrared Einstein response of [1]; it is Lorentzian, smooth in the weak-curvature regime, and reduces to $2\eta^{\mu\nu}$ at spatial infinity.

The flat-space Casimir value is $A_\tau^\infty = 2$ (Q11 [5]); we use this as the reference value for the temporal component.

2.2 Fibrewise BI sphere

Definition 2.1 (Local capacity radius)**.** *Let $\epsilon(x) \geq 0$ be the* local spectral occupancy *at $x \in M$: the fraction of the BI capacity budget at x already occupied by the local projected configuration. The* local capacity radius *is*

$$R(x)^2 := 1 - \epsilon(x). \tag{5}$$

In the homogeneous vacuum ($\epsilon \equiv 0$) one recovers $R(x) = 1$ and the unit Born–Infeld sphere $(F_n^\tau)^2 + B_n^2 = 1$.

Definition 2.2 (Local normalised mobility and fibrewise filling law)**.** *The* local normalised spatial mobility $\beta(x,n) \in [0,1]$ *is defined by* $B(x,n) = R(x)\,\beta(x,n)$*, where* $B(x,n)$ *is the O7 spatial coverage fraction at position* x *and depth* n*. The* local Weil filling variable *is* $f(x,n) := 1-\beta(x,n)$*.*

Remark 2.3 (Fibrewise filling law)**.** *The Weil factorization argument of [3] (Corollary 1) gives* $\Sigma_n^{\mathrm{tot}} = 1 - B_n$ *globally. In the fibrewise setting, the Weil representation is local to each BFS history, and the same argument yields*

$$\Sigma^{\mathrm{tot}}(x,n) = R(x)\,\big(1-\beta(x,n)\big) = R(x)\,f(x,n), \tag{6}$$

i.e. the total residual is the local budget fraction not yet reached by the spatial BFS. The linear filling law holds in the normalised variable $f = 1-\beta$*:* $\beta(x,n) + f(x,n) = 1$*.*

Definition 2.4 (Fibrewise capacity flow vector)**.** *The* local capacity flow vector *at* (x,n) *is*

$$\vec{F}(x,n) = F^{\tau}(x,n)\,\partial_{\tau} + \vec{F}^{\,\mathrm{spat}}(x,n) \;\in\; T_xM,$$

with $|\vec{F}^{\,\mathrm{spat}}(x,n)| = B(x,n)/dn = R(x)\beta(x,n)$*. The BI budget at* x *places* $\vec{F}(x,n)$ *on the* local BI sphere*:*

$$\big(F^{\tau}(x,n)\big)^2 + \big(R(x)\beta(x,n)\big)^2 = R(x)^2. \tag{7}$$

3 Spectral Occupancy and Metric Identification of $R(x)$

3.1 Local spectral occupancy from the admissibility data

In the Cosmochrony framework, a localised mass-energy configuration is a projected configuration that has depleted part of the BI capacity budget in a neighbourhood of x. More precisely, the admissible capacity budget at x is the fraction of the spectral space $\mathrm{Sym}^2(V_\rho)$ available for further projective transitions:

$$\epsilon(x) = \frac{\sigma_{\mathrm{loc}}(x)}{\sigma_{\max}}, \tag{8}$$

where $\sigma_{\mathrm{loc}}(x)$ is the spectral capacity already occupied by the configuration at x and $\sigma_{\max} = c_{\mathrm{BI}}^2$ is the maximal budget. In the homogeneous vacuum, $\sigma_{\mathrm{loc}} \equiv 0$ and $R(x) = 1$.

3.2 Identification with the effective co-metric

The completed effective co-metric at x is $g^{\mu\nu}(x)$. Its temporal component satisfies $g^{\tau\tau}(x) < 0$ (Lorentzian signature). The flat-space reference value is $g^{\tau\tau,\infty} = -A_\tau^\infty = -2$ (Q11 [5]).

Definition 3.1 (Metric capacity lapse)**.** *The* metric capacity lapse *at* x *is the ratio of the local temporal metric component to its flat-space reference value:*

$$R(x)^2 := \frac{g_{\tau\tau}(x)}{g_{\tau\tau}^{\infty}} = -2\,g_{\tau\tau}(x), \tag{9}$$

where $g_{\tau\tau}(x) = (g^{\tau\tau}(x))^{-1}$ *is the inverse metric temporal component,* $g_{\tau\tau}^{\infty} = -1/2$ *is its flat-space reference value (from* $g^{\mu\nu} = 2\eta^{\mu\nu}$*, Q11 [5]), and the explicit form* $-2g_{\tau\tau}(x)$ *follows from* $g_{\tau\tau}^{\infty} = -1/2$*. In the homogeneous vacuum* $g_{\tau\tau}^{\infty} = -1/2$*, giving* $R(\infty) = 1$*.*

Remark 3.2 (Postulated equivalence of the two definitions)**.** *Definitions 2.1 and 3.1 are postulated to describe the same local capacity depletion in the Einstein regime. Their direct equivalence from the admissibility data alone is not proved here; establishing* $\epsilon(x) = 1 - R(x)^2 = 1 + 2g_{\tau\tau}(x)$ *purely from spectral occupancy is stated as open problem LC-O2-O1. The metric formulation* (9) *is used for the weak-field derivation in Section 6.*

4 Fibrewise Temporal Residual Map

4.1 The Lorentzian projector at x

The local co-metric $g^{\mu\nu}(x)$ determines a Lorentzian temporal projector $P_{\tau,x}$ on T_xM by the same formula as in the homogeneous case ([4], Section 4.1):

$$P_{\tau,x}\,V = V^\tau\,\partial_\tau, \qquad V \in T_xM. \tag{10}$$

4.2 Fibrewise temporal residual map

Definition 4.1 (Fibrewise temporal residual map)**.** *The* fibrewise temporal residual map $\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}$ *is the composition:*

$$\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\colon \Sigma^{\mathrm{tot}}(x,n) \xrightarrow{\textit{BI lift at } x} \vec{F}(x,n) \xrightarrow{P_{\tau,x}} F^\tau(x,n)\,\partial_\tau \xrightarrow{\textit{magnitude}} |F^\tau(x,n)|.$$

4.3 Main theorem

Theorem 4.2 (Fibrewise temporal residual map)**.** *Assume the Q5a continuum hypotheses [2], the spectral universality* [U] *proved in U1 [6], the fibrewise BI capacity lift of Definition 2.4, and the local co-metric $g^{\mu\nu}(x)$ of the effective Lorentzian geometry. Then the fibrewise temporal residual map acts by*

$$\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\,\Sigma^{\mathrm{tot}}(x,n) \;=\; R(x)\,\sqrt{f(x,n)\big(2-f(x,n)\big)}, \tag{11}$$

where $f(x,n) = 1-\beta(x,n)$, $R(x)$ is the local capacity radius (9)*, and $\beta(x,n)\in[0,1]$ is the local normalised spatial mobility.*

The quadratic residual law holds fibrewise:

$$\big(R(x)\beta(x,n)\big)^2 + \big(\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\,\Sigma^{\mathrm{tot}}(x,n)\big)^2 = R(x)^2. \tag{12}$$

Moreover, $\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}$ is the unique positive function satisfying simultaneously:

(L_x) $\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\Sigma^{\mathrm{tot}}(x,n) = |P_{\tau,x}\vec{F}(x,n)|$ *(local Lorentzian projection);*

(BI_x) $B(x,n)^2 + (\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\Sigma^{\mathrm{tot}}(x,n))^2 = R(x)^2$ *(local BI sphere* (7)*);*

(W_x) $f(x,n) = 1-\beta(x,n)$ *with $B(x,n) = R(x)\beta(x,n)$ (fibrewise filling law* (6)*).*

Proof. **Explicit formula.** From (BI_x): $(F^\tau)^2 = R(x)^2 - B(x,n)^2 = R(x)^2 - R(x)^2\beta^2 = R(x)^2(1-\beta^2)$. Hence $|F^\tau| = R(x)\sqrt{1-\beta^2}$. By ($\mathrm{W}_x$): $\beta = 1-f$, so $1-\beta^2 = 1-(1-f)^2 = f(2-f)$. Therefore

$$\mathcal{P}^{\mathrm{Q5b}}_{\tau,x}\Sigma^{\mathrm{tot}}(x,n) = R(x)\sqrt{f(x,n)(2-f(x,n))}.$$

Equation (12) follows: $R^2\beta^2 + R^2(1-\beta^2) = R^2$.

Uniqueness. Let $\psi_x(f)$ satisfy (L_x), (BI_x), (W_x). From (W_x): $B(x,n) = R(x)(1-f)$. Substituting into (BI_x): $R(x)^2(1-f)^2 + \psi_x(f)^2 = R(x)^2$, giving $\psi_x(f) = R(x)\sqrt{f(2-f)}$. The positive branch is uniquely forced.

Factorisation. The formula $R(x)\sqrt{f(2-f)}$ separates into a local factor $R(x)$ depending only on position and a universal factor $\sqrt{f(2-f)}$ which is identical to the homogeneous case of [4]. This factorisation (3) is not postulated; it follows from the normalised-capacity definition of β. □

Corollary 4.3 (Homogeneous case)**.** *Setting $R(x)\equiv 1$ (vacuum, $\epsilon\equiv 0$) in Theorem 4.2 recovers the flat-space result of [4]:*

$$\mathcal{P}^{\mathrm{Q5b}}_{\tau}\Sigma^{\mathrm{tot}}_n = \sqrt{1-B_n^2}, \qquad B_n^2 + \big(\mathcal{P}^{\mathrm{Q5b}}_{\tau}\Sigma^{\mathrm{tot}}_n\big)^2 = 1.$$

Proof. Immediate from Theorem 4.2 with $R\equiv 1$: $1\cdot\sqrt{f(2-f)} = \sqrt{(1-B_n)(1+B_n)} = \sqrt{1-B_n^2}$. □

5 Gravitational Time Dilation

Corollary 5.1 (Fibrewise Lorentz factor and gravitational time dilation)**.** *Under the hypotheses of Theorem 4.2, the fibrewise temporally projected saturation ratio is*

$$\eta_x^\tau := \frac{B(x,n)}{\mathcal{P}_{\tau,x}^{\mathrm{Q5b}}\Sigma^{\mathrm{tot}}(x,n)} = \frac{R(x)\beta}{R(x)\sqrt{1-\beta^2}} = \beta\gamma, \tag{13}$$

independent of $R(x)$, where $\gamma = (1-\beta^2)^{-1/2}$. Consequently $\Phi(\eta_x^\tau) = 1/\gamma$ as in the homogeneous case.

The temporal residual per unit cascade depth at position x is

$$\left.\frac{d\tau}{dn}\right|_x = |F^\tau(x,n)| = R(x)\,\frac{1}{\gamma} = \frac{R(x)}{\gamma}. \tag{14}$$

Gravitational time dilation is the factor $R(x) \le 1$ multiplying the special-relativistic dilation $1/\gamma$:

$$\frac{(d\tau/dn)|_x}{(d\tau/dn)|_\infty} = \frac{R(x)/\gamma}{1/\gamma} = R(x). \tag{15}$$

Near a massive body $R(x) < 1$, and clocks at position x run slower than clocks at infinity by the factor $R(x)$.

Proof. The ratio η_x^τ in (13) cancels $R(x)$ in numerator and denominator. Equation (14) is $|F^\tau| = R(x)\sqrt{1-\beta^2} = R(x)/\gamma$. Equation (15) follows by comparison with the flat-space value $R(\infty) = 1$. □

Remark 5.2 (Light cone at x)**.** *$\beta \to 1$ (maximal spatial mobility) forces $F^\tau \to 0$ for any $R(x) > 0$: the light cone condition is local and independent of the gravitational field. At x with $R(x) = 0$ (if such a point existed), all temporal capacity is exhausted by the local configuration; this would correspond to a horizon-like saturation locus or to a breakdown of the effective metric description.*

6 Weak-Field Limit

6.1 Connecting $R(x)$ to $g_{\tau\tau}(x)$

From Definition 3.1: $R(x)^2 = -2\,g_{\tau\tau}(x)$. The effective co-metric $g^{\mu\nu}(x)$ is determined by the infrared Einstein response of [1]: the renormalized spectral entropy variation gives $\delta S_\Pi^{\mathrm{ren}}/\delta g^{\mu\nu} \propto G_{\mu\nu}$, leading to the effective Einstein equations.

In the static weak-field (Newtonian) regime, the Einstein equations give

$$g_{\tau\tau}(x) \;\approx\; -\tfrac{1}{2}\Big(1 + \frac{2\Phi_N(x)}{c^2}\Big), \tag{16}$$

where $\Phi_N(x)$ is the Newtonian potential satisfying $\nabla^2\Phi_N = 4\pi G_N\rho$. The $1/r$ profile of Φ_N arises as the Green function of the Laplace-type operator A_g of [1]: this is the same mechanism as the Newtonian interaction emerging from the spectral resolvent (Section 2.3 of [1]).

6.2 Weak-field result

Corollary 6.1 (Weak-field gravitational time dilation)**.** *Under the hypotheses of Theorem 4.2 and in the weak-field regime $|\Phi_N(x)| \ll c^2$, the local capacity radius satisfies*

$$R(x) = \sqrt{-2\,g_{\tau\tau}(x)} \;\approx\; 1 + \frac{\Phi_N(x)}{c^2}, \tag{17}$$

and the gravitational time dilation formula (14) *becomes*

$$\left.\frac{d\tau}{dn}\right|_x \approx \left(1+\frac{\Phi_N(x)}{c^2}\right)\frac{1}{\gamma}, \tag{18}$$

in agreement with the standard result of general relativity at leading weak-field order.

Proof. In the Q5b/Q11 convention ($g^{\mu\nu} = 2\eta^{\mu\nu}$), the flat-space inverse metric component is $g^{\infty}_{\tau\tau} = -1/2$. The weak-field Einstein equations of [1] give, for a static Newtonian potential, $g_{\tau\tau}(x) \approx -\frac{1}{2}\big(1 + 2\Phi_N(x)/c^2\big)$. Substituting into (9):

$$R(x)^2 = -2\, g_{\tau\tau}(x) = -2\cdot\left(-\tfrac{1}{2}\right)\big(1+2\Phi_N/c^2\big) = 1 + \frac{2\Phi_N(x)}{c^2}.$$

At first order in Φ_N/c^2: $R(x) \approx 1 + \Phi_N(x)/c^2$. Equation (18) follows from (14). The 1/r profile of $\Phi_N(x)$ is the Green function of A_g in three spatial dimensions, as established in [1] Section 2.3. □

Remark 6.2 (Status of the weak-field result)**.** *Corollary 6.1 is conditional on two imports: (i) the effective Einstein equations from [1] (in the infrared, weak-curvature, regime $R\ell_{\mathrm{sp}}^2 \ll 1$); (ii) the identification of $\Phi_N(x)$ as the Newton coupling via the Green function of A_g (Section 2.3 of [1]). No new hypothesis is introduced here. The result is a consequence of combining the capacity-lapse language with the gravitational output of [1].*

7 Logical Status and Open Problems

7.1 Status table

Label	Content	Status
[4] Thm. 4.5	Homogeneous map $\sqrt{1-B_n^2}$, uniqueness	Proved (under [U])
Q11 [5]	$A_\tau = 2$, flat metric $g^{\mu\nu} = 2\eta^{\mu\nu}$	Proved (under [U])
Definition 2.1	$R(x)^2 = 1 - \epsilon(x)$	Structural (this paper)
Definition 3.1	$R(x)^2 = -2g_{\tau\tau}(x)$	Structural (this paper)
Theorem 4.2	Fibrewise map, uniqueness	Proved (under [U] + fibrewise BI lift)
Corollary 5.1	$d\tau/dn\|_x = R(x)/\gamma$	Proved (under [U] + fibrewise BI lift)
Corollary 6.1	$R(x) \approx 1 + \Phi_N/c^2$	Conditional (imports [1])

7.2 What is not proved here

The present paper does not derive the Einstein field equations. That derivation is in [1]. The present paper does not derive $R(x)$ from first principles as a function of the spectral occupancy $\epsilon(x)$. It gives two definitions (Definitions 2.1 and 3.1) that are equivalent in the Einstein regime, but the bridge $\epsilon(x) \leftrightarrow g_{\tau\tau}(x)$ is not closed analytically within the capacity-flow language alone. Closing this bridge is LC-O2's remaining open problem (LC-O2-O1 below).

7.3 Open problems

LC-O2-O1 (Spectral closure of $R(x) \leftrightarrow g_{\tau\tau}(x)$). Definitions 2.1 and 3.1 give two expressions for $R(x)$: one from spectral occupancy $\epsilon(x)$ and one from the effective metric. A direct spectral derivation showing

$$\epsilon(x) = 1 - R(x)^2 = 1 + 2g_{\tau\tau}(x)$$

from the admissibility data alone would close this identification without importing the Einstein equations. This is the capacity-flow analogue of deriving gravitational time dilation from the spectral operator structure directly.

LC-O2-O2 (Length contraction in the fibrewise regime). Corollary 5.1 gives fibrewise time dilation and the light cone. Length contraction in the non-homogeneous setting requires a comparison of spatial capacity across different fibres and the reconstruction of effective Lorentz transformations between locally inertial observers in the curved geometry.

LC-O2-O3 (Strong-field regime: $R(x) \to 0$). Remark 5.2 identifies $R(x) = 0$ with a horizon-like saturation locus or a breakdown of the effective metric description. The capacity-flow language provides a novel criterion for horizon formation: the local capacity budget is fully saturated by the configuration. Whether this criterion reproduces the Schwarzschild horizon condition is open.

Acknowledgements

The author acknowledges the use of large language models as a supportive tool for refining language, structure, and internal consistency during the development of this manuscript.

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